

JOB DESCRIPTION

JOB DETAILS

Job Title:	Band 5: Healthcare Science Practitioner (Clinical Technologist)
Department / Ward:	Diagnostic Radiology and Radiation Protection Service (DRRPS) within the Imaging Physics and Radiation Protection (IPRP) Group – a regional NHS physics service for the North West.
Division:	Christie Medical Physics & Engineering (CMPE)
Base:	The Christie, Withington. However, due to the large regional physics service and North West Imaging Networks formation, this post requires regular travel and customer location working throughout North West and infrequently, further afield. Some working from home for non-clinical work is satisfactory and will likely be part of on-going work patterns where this does not impact negatively on service delivery and team cohesion. There is potential, once trained, to be based in the Lancashire and South Cumbria Integrated Care System/Imaging Network for those who live further north than Manchester.

ORGANISATIONAL ARRANGEMENTS

Accountable to:	1. (Managerially & Professionally) Group Leader, Imaging Physics and Radiation Protection Group 2. (Reporting) Physicist or Senior Technologist, Diagnostic Radiology & Radiation Protection Service
Other Accountabilities:	1. (Professionally) Lead Healthcare Scientist 2. (Clinically) Clinical Directors of Radiology in organisations where CMPE provide services
Responsible for:	1. (Supervision) Healthcare Science Assistants, Associates and Practitioners – in training 2. (Supervision) Clinical Scientists in training when experienced post holder has competence in the skill set and can cascade the knowledge



JOB PURPOSE

To assess the safety and performance of X-ray imaging equipment, assist in the provision of technical support, radiation protection and patient dosimetry advice to users of ionising radiations, in order to optimise the use of X-rays in diagnostic imaging and comply with regulatory requirements. Activities fall within the framework provided by a Medical Physics Expert or Radiation Protection Adviser.

Please note that this role is in diagnostic radiology physics, not radiotherapy or nuclear medicine. It is not a 'radiography post' nor patient-facing. However, radiographers with experience of performing QA on diagnostic equipment hoping for a change of field would be suitable candidates.

DUTIES AND RESPONSIBILITIES

Equipment performance testing and radiation protection surveys (main focus of role)

- Undertakes equipment performance and radiation protection surveys, alone or with a colleague, following standard operating procedures defined within an ISO9001 quality system:
 - Performs routine quality control on existing installations
 - Assists in the critical examination and commissioning of new equipment to ensure that it is safe for bringing into clinical use
 - Assists with investigation of faults on existing installations
 - Calibrates patient dose measuring equipment to confirm the accuracy of recorded doses
 - Checking the radiation protection of the installation to ensure staff and public safety
- This will involve accurate positioning and methodical use of test objects.
- Analysing the results of equipment performance measurements
- Write electronic reports for the Radiation Protection Advisor/Medical Physics Expert and customers, detailing findings and recommending actions required
- Calibrating x-ray dose measuring equipment
- Perform testing of UV units, as required by the group.
- providing advice on level A QC testing, review results and report findings.

Radiation Protection

- Assist in providing radiation protection advice and support as required in the team. This includes
 - producing ad hoc reports on equipment or customer sites, e.g. lists of equipment due, inter-comparison of similar equipment types
 - updating central records of measurements, e.g. typical patient doses, equipment calibration factors
 - calibrating radiation protection meters as required and/or organising dispatch of equipment for external calibration.

Supervision and responsibilities for information resources



- Maintains accurate records using Excel and Access (or other databases as rolled out in the NHS), e.g. results from testing, details of equipment
- Liaise with radiographic staff to book survey visits in conjunction with the Booking Team as required
- Supervise less experienced healthcare science staff (assistants, associates, practitioners or scientists) when undertaking equipment performance and radiation protection surveys as part of their training.
- Undertake internal audits.
- A take a lead on a number of group functions such as H&S checks, oversight of online QA platforms, maintaining our inventory etc, as required

Professional

- Undertakes training identified as required for role, including continuing professional development
- Gains and maintains registration as clinical technologist
- Follow the appropriate key work instructions, risk assessments and systems of work and ensure direct supervision of those you are responsible for until they are competent.
- Collaborates with other members of the group in the improvement of test protocols.
- Contributes to the teaching (and outreach) commitments of the service such as assisting in the preparation and delivery of material, e.g. at the Radiation Protection Supervisors courses.
- Contributes to the research and development of the service, e.g. by carrying out measurements or processing data

Note:

This job description indicates the duties and responsibilities that are appropriate to meet the present needs of the service. Since from time to time these needs may change it is necessary to recognise that the post holder must be willing to undertake other and/or different duties that may, after discussion, be assigned by the Director.

Date Prepared: 7/8/24

Prepared By: Will Mairs

Agreed By:

Employee's Name and Signature:

Date:

Manager's Name and Signature:

Date:

Date Reviewed:

Reviewed By:



OUR VALUES AND BEHAVIOURS



Our Values and Behaviours shape the way we work. Values are beliefs or principles that are important and meaningful to us – they are what drives us. Behaviours are observable actions that bring these values to life. Our behaviours demonstrate our values through how we do things, what we say, how we say it, and how we treat others and expect to be treated ourselves.



Job Title: **Clinical Technologist (Diagnostic Radiology)**

	<u>ESSENTIAL</u>	<u>DESIRABLE</u>	METHOD OF ASSESSMENT
QUALIFICATIONS	- GCSE English*, Maths and ICT or science (C grade & above) - Degree or equivalent in appropriate STEM subject - Eligible for registration with the Register of Clinical Technologists or HCPC**	Registration in Diagnostic Radiology / Rad. Prot. field	Document check
EXPERIENCE	- Operation of diagnostic X-ray equipment across a range of modalities (independently) - QA of rad/CR/DR equipment - QA of fluoroscopy mobiles - QA of dental equipment - Writing equipment performance reports for QA visits - Performing investigations into DX X-ray equipment performance issues	- QA of more advanced x-ray equipment - RP tasks (see notes in JD) - Calibration of dosimeters - Providing training or advice for Level A (radiographer) QA - Assisting with non-routine QA of diagnostic X-ray equipment - Supervision/training of staff	Application form Interview References
SKILLS	- High level of literacy and numeracy - Ability to operate complex imaging equipment - Ability to position tests tools with precision, routinely - Ability to operate radiation instruments/dosimeters - Proficient in using Microsoft Office, scientific software and word processing to analyse data and produce reports - Ability to problem solve/fault find/investigate issues - Ability to work independently and as part of a team - Ability to analyse complex scientific data and clearly communicate findings/recommendations - Ability to apply health & safety and	- Advanced Excel skills - Training / presenting	Application form Interview References



	- risk management principles to practical scenarios - Ability to handle challenging situations with professionalism - Good interpersonal and communication skills		
KNOWLEDGE	- Working knowledge of appropriate UK radiation legislation and practical radiation safety principles - Knowledge of radiation physics and a wide range of diagnostic X-ray equipment - Working knowledge of radiation instrumentation and test objects - Working knowledge of recommended standards of X-ray equipment - Knowledge of X-ray equipment life cycle and stages of QA	Basic knowledge of quality systems	Application form Interview References
VALUES	- Ability to demonstrate the organisation's values and behaviours including respect towards others at all times. - Patient focused - Commitment to reflective practice and CPD - Adherence to RCT code of conduct		Application form Interview References
OTHER	- Valid driving licence for manually operated cars. - Willing to work flexibly within group expectations (see notes below) - Physically able to lift heavy measuring equipment (up to 15 kg) - Enthusiasm, integrity and commitment to this role		Application form Interview Document check

Note on current group work patterns



- Willing to travel on frequent basis (~3 times per week) throughout NW England region and infrequently further afield
- Willing to stay overnight infrequently for equipment testing at more distant hospitals
- Willing to work flexibly in line with service needs. Typical hours the team work are between 8am and 6pm Mon-Fri. Due to the nature of the job there are occasions when work is required outside normal hours.
- Infrequent weekend work required, with notice.
- NB: it's likely the team will move towards working longer days when off site in the region to make visits more efficient e.g. 9.5 hour days.

For appointment at Band 4 (typically in order to train for Band 5)

****Training/development will be provided for those without sufficient relevant knowledge, skills and experience (and registration) who are deemed suitable for the post based on their application and interview. They will be appointed on the bottom of NHS Band 4 pay scale. On the job training will allow candidates to develop competence to work independently. When sufficient experience of independent working across a range of modalities is gained, demonstrated through sign off on our internal training matrix (which assesses theory and practice), we will move the employee to the bottom of Band 5 (with or without registration).**

There are formal training schemes that are supported in our group for a suitable candidate. Progression in the career structure of Clinical Technologists benefits from registration and lack of registration eventually introduces a progression ceiling.

- Equivalence route via Register of Clinical Technologists – those with a relevant degree and transferable experience or experience gained in our team)
- IPEM clinical technologist training scheme – for those with a relevant degree. CMPE is an accredited training centre however this is resource heavy for candidates and our team of trainers, so is not suited to all.

***Ability to use English (based on IPEM requirements)**

The ability to communicate effectively in English is critical to working effectively as a health professional in the UK. For entry onto the training scheme the following will apply regardless of the training route:

- a. Any trainee whose first language is English requires a level 2 qualification (e.g. GCSE at grade C or above) or equivalent.
- b. Any trainee whose first language is not English will be required to obtain an International English Language Testing System (IELTS) certificate. A minimum score of 7.0 with no element below 6.5 is required. If any other testing system is used, you must be able to provide equivalence with IELTS.
- c. Any overseas (non-UK) qualifications must demonstrate equivalence comparison using the UK NARIC (National Academic Recognition Information Centre) service.

PTO



	<u>ESSENTIAL</u>	<u>DESIRABLE</u> <u>(all receive enhanced scoring if gained in the field of medical physics / radiation protection / imaging)</u>
QUALIFICATIONS	- GCSE English*, Maths and ICT or science (C grade & above) - Degree or equivalent in appropriate STEM subject	Qualifications that will directly transfer to gaining registration as a clinical technologist (e.g. A-level in Biology)
EXPERIENCE		- Experience as a clinical technologist in another discipline - Experience working with/managing equipment, testing regimes, QA, Health & Safety, standard operating procedures - Experience in the field of radiography. - MS Office computer skills e.g. Word, Excel - Familiarity with use of databases - Supervision
SKILLS	<i>As above but do not need: 'operating imaging equipment and radiation instrumentation'</i>	Teaching / training / presenting
KNOWLEDGE		- Quality systems - Working knowledge of UK standards and legislation in a particular field
VALUES, OTHER	<i>As above</i>	



GENERAL STATEMENTS:

RISK MANAGEMENT

It is a standard element of the role and responsibility of all staff of the Trust that they fulfil a proactive role towards the management of risk in all of their actions. This entails the risk assessment of all situations, the taking of appropriate actions and reporting of all incidents, near misses and hazards.

RECORDS MANAGEMENT/DATA PROTECTION

As an employee of the Trust, you have a legal responsibility for all records (including patient health, financial, personal and administrative) that you gather or use as part of your work within the trust. The records may be paper, electronic, microfiche, audio or videotapes, x-ray images. You must consult your manager if you have any doubt as to the correct management of the records with which you work.

HEALTH AND SAFETY REQUIREMENTS

All employees of the Trust have a statutory duty of care for their own personal safety and that of others who may be affected by their acts or omissions. Employees are required to co-operate with management to enable the Trust to meet its own legal duties and to report any circumstances that may compromise the health, safety and welfare of those affected by the Trust undertakings.

CONFIDENTIALITY AND INFORMATION SECURITY

As a Trust employee you are required to uphold the confidentiality of all records held by the trust, whether patient records or trust information. This duty lasts indefinitely and will continue after you leave the trust employment.

All Information which identifies individuals in whatever form (paper/pictures, electronic data/images or voice) is covered by the Data Protection Act 2018 and should be managed in accordance with this legislation.

SAFEGUARDING

All employees of The Christie NHS Foundation Trust have a responsibility to safeguard children, young people, and vulnerable adults, regardless of their role. Appropriate and proportionate actions/interventions should be undertaken, documented, and escalated promptly to the Safeguarding Team within normal working hours, or the Duty Manager out of hours. This includes making safeguarding referrals, risk assessments, and incident reports, in-line with roles and responsibilities.

TRUST POLICIES



The Trust operates a range of policies, e.g. Human Resources, Clinical Practice (available on the Trust intranet). All Trust employees must observe and adhere to the provisions outlined in these policies.

EQUALITY, DIVERSITY AND INCLUSION

The Christie NHS Foundation Trust is committed to advancing equality, diversity and inclusion for all our patients, other service users and staff. We want to ensure that everyone who works at the Christie or uses our services is welcomed, valued and treated with dignity and respect.

It is your responsibility to understand and work in line with the Trust's equality, diversity, inclusion and human rights policies. You should value others and treat everyone you come into contact with at work with fairness, dignity and respect at all times and uphold their human and other rights.

INFECTION CONTROL

Healthcare workers have an overriding duty of care to patients and are expected to comply fully with the best practice standards. You have a responsibility to comply with Trust policies for personal and patient safety and for prevention of healthcare-associated infection (HCAI); this includes a requirement for rigorous and consistent compliance with Trust policies for hand hygiene, use of personal protective equipment and safe disposal of sharps. Knowledge, skills and behaviour in the workplace should reflect this; at annual appraisal you will be asked about application of practice measures known to be effective in reducing HCAI

ENVIRONMENTAL SUSTAINABILITY

All employees of the Trust have a responsibility to ensure they have an awareness of environmental sustainability issues which affect the Trust and to contribute to the achievement of the reduction of the Trust's environmental and energy performance footprint e.g. (but not limited to) the use of energy consumed in workspaces (heat/light/paper consumed) and to recycle consumable products wherever possible using appropriate facilities.

